

Event sequences and networks

part 2. Co-presence networks

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The peculiarity of politics in Slovenia

- In *four* penultimate elections in Slovenia, the single biggest (wining) party was the one that was formed less than 6 months before the elections.
- Therefore, the voters picked the future representatives also who were not active in the political sphere but several of them are known because of their professional careers (e.g., journalists, university professors).



Research question

Do the former connections in different areas affect political connections among the current political office holders?

- If two people have *collaborated* in several different situations in the past, *trust* can develop between them. Because of this trust, someone who *chooses her/his colleague in the political sphere* may prefer to choose this trustworthy person rather than someone with whom (s)he has had no relations in the past.
- As *social capital* is a set of resources – such as trust or information – contained in people's networks (J. Coleman, 1988) the research question is:

Does the social capital of a political office holder (e.g. Prime Minister, minister) affect the selection of his/her colleagues in the government?

James Coleman (1988). Social capital in the creation of human capital. American Journal of Sociology, 94: 95-120.



Oštro, Zvezoskop, data

- *Oštro* is the Center for investigative journalism in the Adriatic region and cultivates investigative and data journalism.
- Colleagues from Oštro have developed the *Zvezoskop* project, which allows the public to independently research the connections between *current ministers, state secretaries and members of parliament*, and their educational and career paths. The careers of 160 politicians are included in Zvezoskop.
- Every piece of information in Zvezoskop is based on primary sources (official documents and data, e.g., CVs), authentic information, as well as confirmations from individual officials.
- If the information could not be placed reliably in time, it was not included in the database, since the period of presence in an institution is important for determining the co-presence between two political officials.



Data

The politicians can be active in various types of activities, which are work, education, free time, political party, and counseling and supervision.

These types of activities define 5 relations.

In the analysis that follows, all types of activities are combined into one, although we also analyzed individual relations.

For each pair of politicians, based on their CVs and additional information, the time (in days) when the two politicians were *simultaneously present* in the same institution and type of activity is calculated (it is a possibility of cooperation, since *co-presence* does not yet mean cooperation, but it is a condition for it) .



Raw data


```
,person_id,person_name,part_of_cv,show_in_network,start_day,start_month,
start_year,institution_si,institution_standardized_si,institution_department_si,
affiliation_type_si,position_si,end_day,end_month,end_year,notes_institution_si,notes_position_s
...
2241,112,Matej Tonin,izobraževanje,True,,9,1998,gimnazija in srednja šola rudolfa maistra kamnik
2242,112,Matej Tonin,izobraževanje,True,,10,2002,ul fakulteta za družbene vede,,,,dodiplomski št
2243,112,Matej Tonin,izobraževanje,True,,10,2007,ul fakulteta za družbene vede,,,,magistrski štu
2244,112,Matej Tonin,delovne izkušnje,True,16/4/2007,,2007,državni zbor,,poslanska skupina NSi,j
2245,112,Matej Tonin,strankarska pozicija,True,,,2006,nsi,,občinski odbor Kamnik,,predsednik,,,2
2246,112,Matej Tonin,strankarska pozicija,True,10/9/2010,,2010,nsi,,,,kanididat za župana v obči
2247,112,Matej Tonin,delovne izkušnje,True,29/11/2006,,2006,občina kamnik,,,politik,občinski sve
2248,112,Matej Tonin,strankarska pozicija,True,,,2007,nsi,,Mlada Slovenija,,podpredsednik,,,2009
2249,112,Matej Tonin,delovne izkušnje,True,3/11/2008,,2008,vizija agencija za komuniciranje mate
2250,112,Matej Tonin,strankarska pozicija,True,,,2010,nsi,,,,podpredsednik,,,2012,,
2251,112,Matej Tonin,delovne izkušnje,False,21/12/2011,,2011,državni zbor,,,politik,poslanec,13/
2253,112,Matej Tonin,delovne izkušnje,True,22/6/2018,,2018,državni zbor,,,politik,predsednik,23/
2254,112,Matej Tonin,delovne izkušnje,False,13/5/2022,,2022,državni zbor,,,politik,poslanec,,,21
2255,112,Matej Tonin,strankarska pozicija,True,22/8/2008,,2008,nsi,,,,kandidat za poslanca,21/9/
2256,112,Matej Tonin,delovne izkušnje,True,13/3/2020,,2020,ministrstvo za obrambo,,,politik,mini
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2260,112,Matej Tonin,svetovalne in nadzorne funkcije etc.,True,,,2015,inštitut dr janeza evangel
2261,112,Matej Tonin,prostočasne aktivnosti,True,27/11/2006,,2006,športno društvo tuhinj,,,,pred
2262,112,Matej Tonin,strankarska pozicija,False,,12,2001,nsi,,,,član,,,2100,,
2263,112,Matej Tonin,strankarska pozicija,False,1/2/2018,,2018,nsi,,izvršilni odbor,,predsednik,
...
```



Reduced data

The cleaned data table was first reduced to variables considered in the analysis.

- R – row (event) number (index)
- ID – `person_name`: person's name
- s – `start_day|month|year`: start date
- f – `end_day|month|year`: finish date
- S – `part_of_cv`: type of activity
- U – `institution_si`: location (institution, party, organization, etc.)

According to the research question, we considered events starting before the penultimate elections on April 24, 2022.



Events

In terms of event sequence analysis, we are considering events

$$e_R = (ID_R, s_R, f_R, S_R, U_R)$$

partitioned by persons

$$\mathbf{S}_{ID} = \{S_{id} : id \in ID\}$$

To each person $id \in ID$ we assign a temporal quantity $TQ_{id} = (tq_i)$

$$e_i \rightarrow tq_i = [s_i, f_i, (S_i, U_i)]$$

According to the research question, we considered events starting before the *April 24, 2022 elections*.



Co-presence network

We decided to base our analysis on the corresponding *co-presence network* – a weighted (number of days of co-presence on a link) multi-relational (5 areas of S) temporal network $N = (V, L, w, t)$. The set of nodes V consists of 160 politicians.





Co-presence network

Person a and person b are co-present if they are both active in some time period in the same type of activity and institution. Similarity between a and b can be measured by the length of their co-presence intervals.

In preliminary experiments, we found that parallel events (triplets) of the same kind can greatly increase the similarity between persons. One way to reduce this influence is to count parallel events in the same institution between two persons only once.

$$sim(a, b) = \sum_{u \in U} |T_u(a, b)|$$

where

$$T_u(a, b) =$$

$$\{t \in T : \exists\ tq_i \in TQ_a, tq_j \in TQ_b : (t \in [s_i, f_i) \cap [s_j, f_j) \wedge U_i = U_j = u)\}$$



Data – co-presence

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```
   R           ID          s          f         S                    U
  64   Igor Papič 1973-09-01 1981-08-01 education          oš v kočevju
1116 Robert Golob 1979-09-01 1983-08-01 education       gim nova gorica
  65   Igor Papič 1981-09-01 1985-06-01 education      vegova ljubljana
1068 Robert Golob 1983-10-01 1989-09-29 education                ul fe
  66   Igor Papič 1986-10-01 1992-01-01 education                ul fe
1070 Robert Golob 1989-11-15 1994-11-14      work                ul fe
1069 Robert Golob 1992-10-01 1994-10-17 education                ul fe
  67   Igor Papič 1992-10-01 1995-01-01 education                ul fe
1115 Robert Golob 1994-01-01 1994-01-01 education tu georgia atlanta us
  70   Igor Papič 1994-01-01 1996-01-01      work             siemens de
1071 Robert Golob 1994-11-15 1995-10-31      work                ul fe
  69   Igor Papič 1993-03-01 1996-01-01      work                ul fe
  68   Igor Papič 1995-10-01 1998-01-01 education                ul fe
1072 Robert Golob 1995-11-01 1997-09-30      work                ul fe
1073 Robert Golob 1997-10-01 2001-06-30      work                ul fe
1078 Robert Golob 1998-11-01 1999-08-31      work   min za gospodarstvo
  71   Igor Papič 1999-01-01 2007-01-01      work                ul fe
  96   Igor Papič 2000-01-01 2002-06-30      work                ul fe
  94   Igor Papič 2001-01-01 2002-12-31      work                ul fe
...
              u           v          s          f       rel    d   Ru   Rv
1   Robert Golob   Igor Papič 1986-10-01 1989-09-29 education 1094 1068   66
2   Robert Golob   Igor Papič 1993-03-01 1994-11-14      work  623 1070   69
3   Robert Golob   Igor Papič 1992-10-01 1994-10-17 education  746 1069   67
4     Igor Papič Robert Golob 1994-11-15 1995-10-31      work  350   69 1071
5     Igor Papič Robert Golob 1995-11-01 1996-01-01      work   61   69 1072
7   Robert Golob   Igor Papič 1999-01-01 2001-06-30      work  911 1073   71
10  Robert Golob   Igor Papič 2000-01-01 2001-06-30      work  546 1073   96
12  Robert Golob   Igor Papič 2001-01-01 2001-06-30      work  180 1073   94
...
          u           v          s          f    d       rel   test
1 Igor Papič Robert Golob 1993-03-01 1996-01-01 1037      work  ul fe
2 Igor Papič Robert Golob 1999-01-01 2021-01-01 8037      work  ul fe
3 Igor Papič Robert Golob 1992-10-01 1994-10-17  747 education  ul fe
```



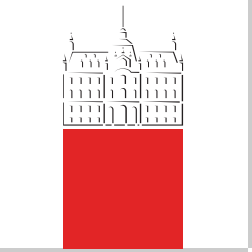
Data

Co-presence networks was constructed for the time before the elections on April 24, 2022.

	relation	number of events
1	work	1767
2	education	470
3	leisure activities	6
4	party	1086
5	counseling & supervision	10
	total	3339
	nodes	154

The link values (number of co-presence days) in the network are in the interval $[1, 3339]$. Applying the square root we transform them into the interval $[1, 58]$. Another application of the square root gives the interval $[1, 8]$, which is used for the visualization.

The analysis was done using program Pajek.



Co-presence network

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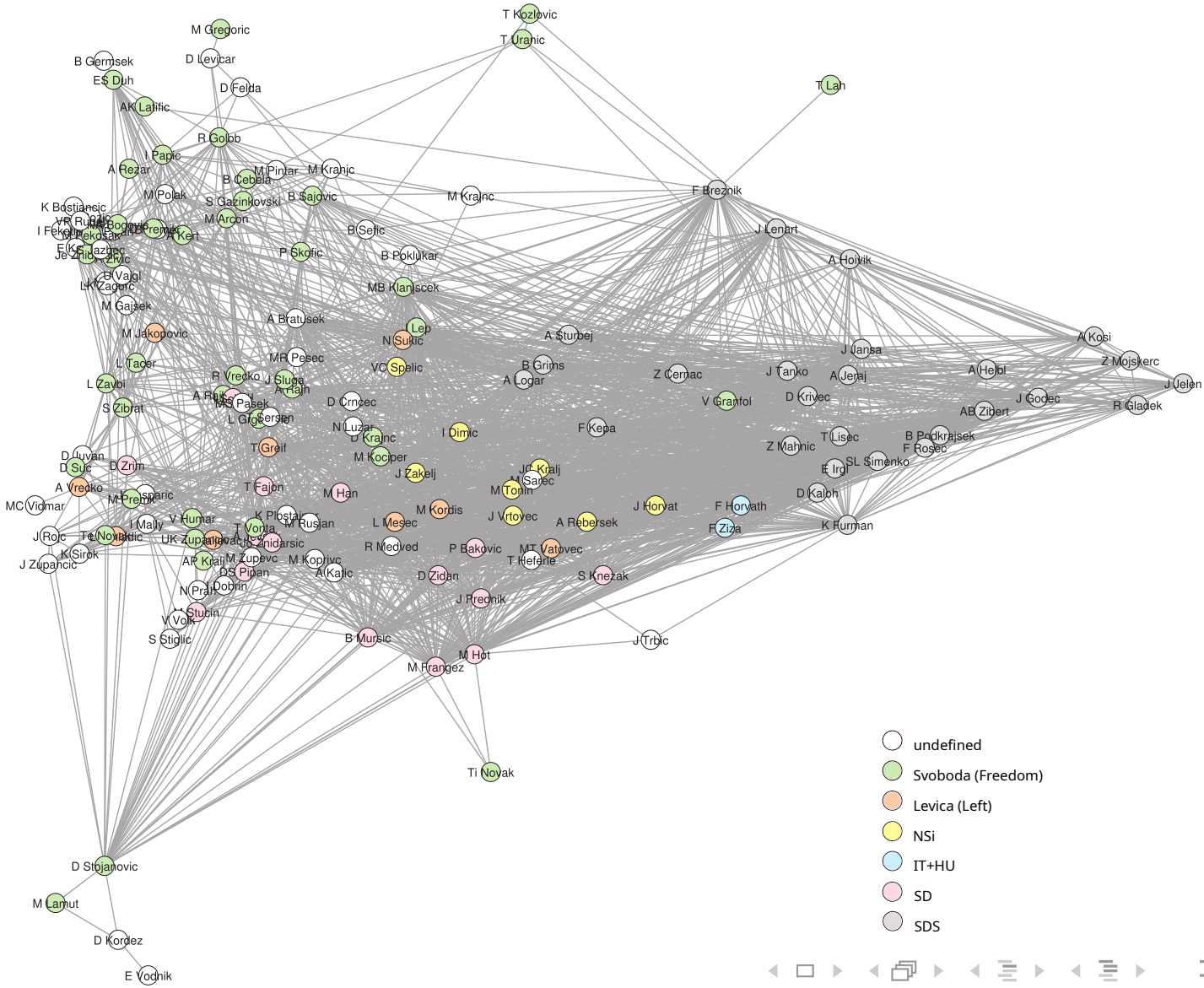
Oštro

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Blockmodeling

The co-presence network is a weighted one. Therefore, we decided to use indirect blockmodeling approach, where

- dissimilarity between nodes was calculated by corrected eucliden distance
- Ward hierarchical agglomerative method was used as a clustering procedure

The obtained dendrogram is presented in the next slide.



Dendrogram

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Research question

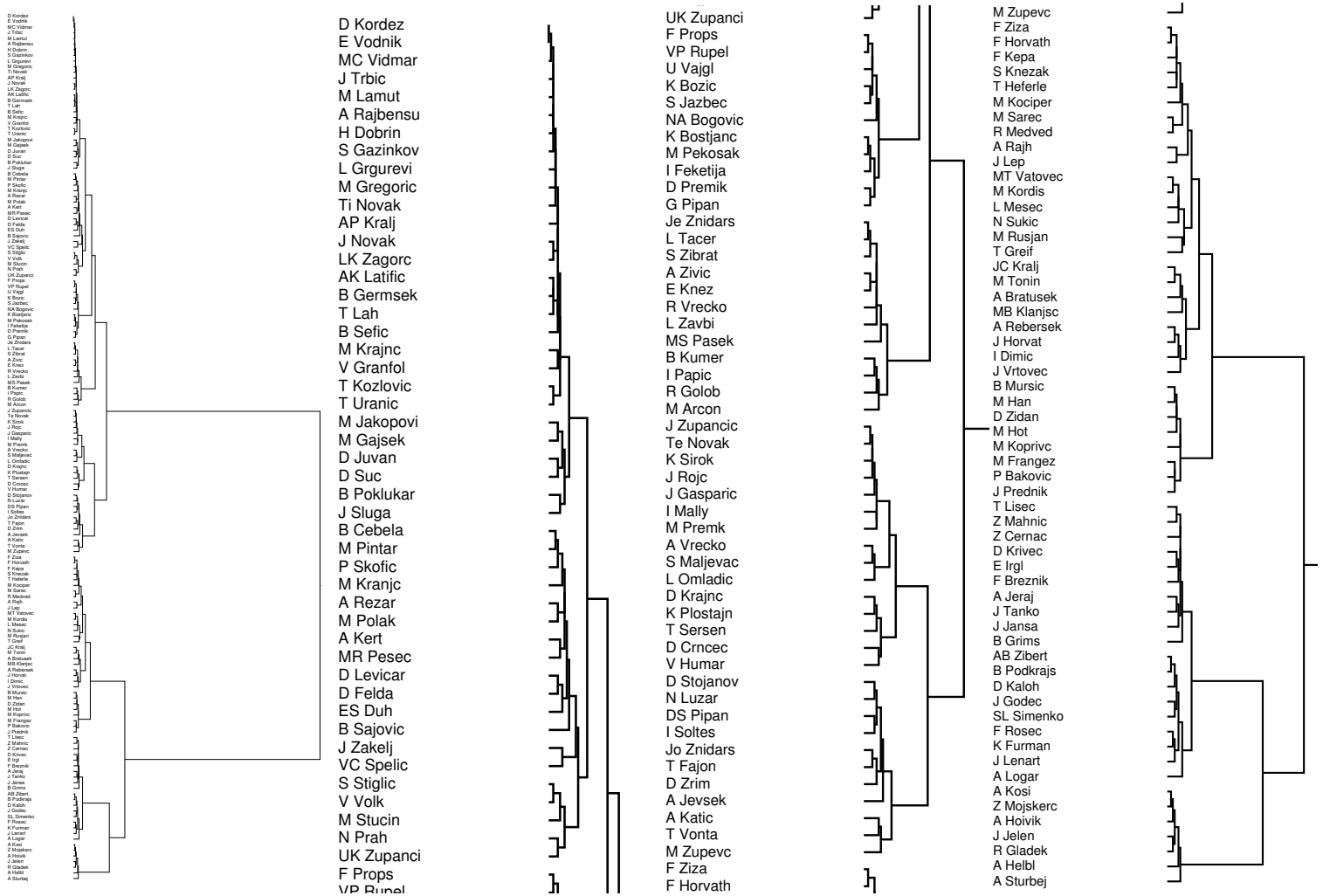
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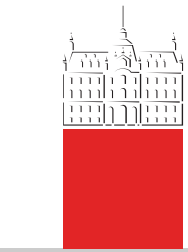


According to the dendrogram 9 clusters of politicians were determined.



Partition

1. **'New' SDS MPs** (7)
2. **'Old' SDS MPs** (19)
3. **'Old' SD MPs** (8)
4. **MPs not members of SDS or SD** (24)
5. **'New' SD MPs** (11)
6. **Members of Gibanje Svoboda (Freedom) and Levica (Left)** (15)
7. **Members of Gibanje Svoboda (Freedom)** (12)
8. **Members of Gibanje Svoboda (Freedom)** (11)
9. **Isolated political office holders** (47)



Reordered network matrix / $\sqrt{w}$

Research question

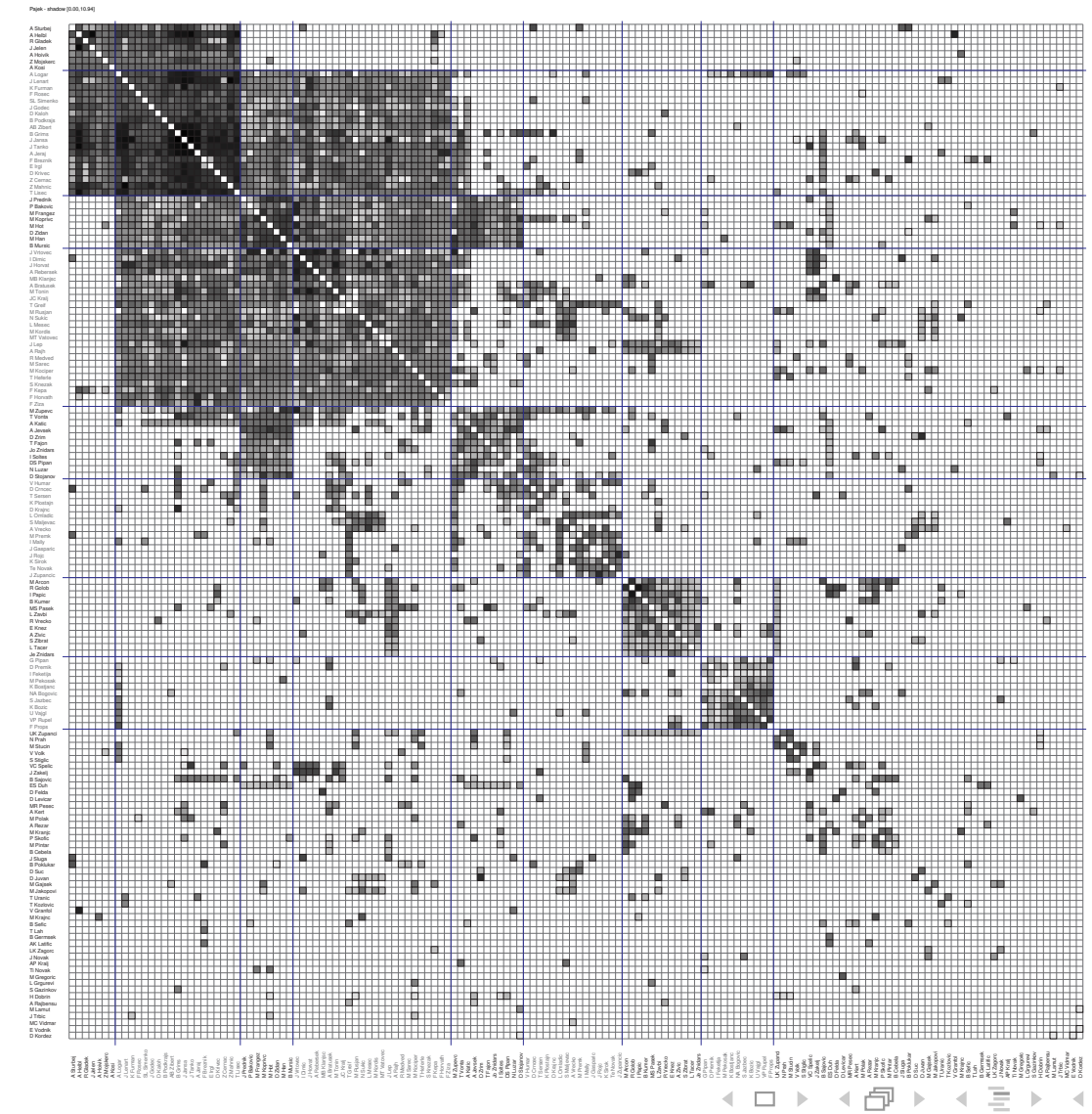
Oštro

Data

Co-presence network

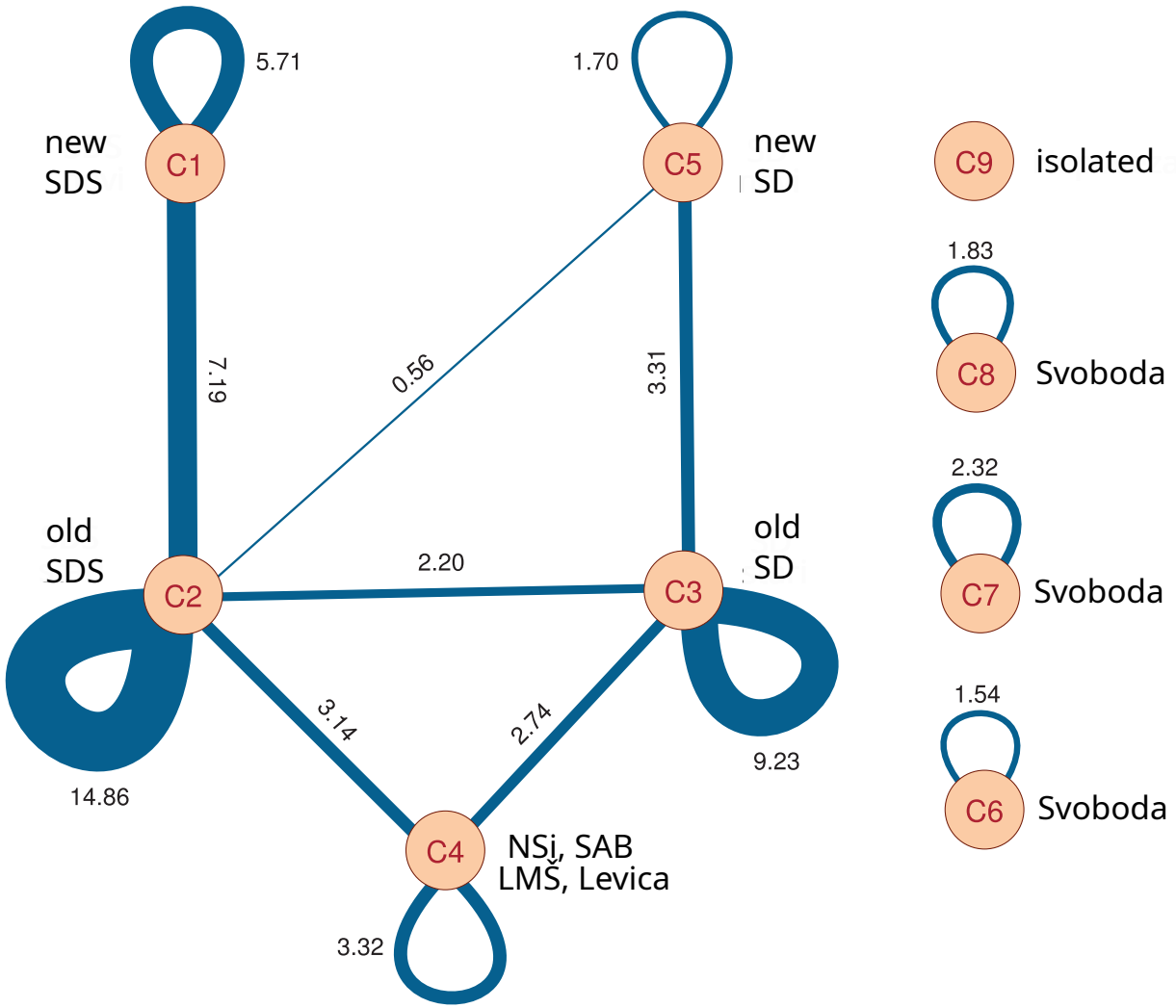
Blockmodeling

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Blockmodel (weights in years)





Some conclusions

- The strongest co-presence is among members of the SDS party (among older members, almost 15 years on average). The members of the second oldest SD party (almost 10 years among the seniors) are also quite connected. The greatest co-presence is expected between the two SDS clusters (C1 and C2) (over seven years on average) and between the SD clusters (C3 and C5) with an average of just over three years.
- The fourth group (C4) consists of politicians who have been active for a long time, but are not members of SDS or SD. They are expected to be co-present together with older SDS members and older SD members (on average around three years).
- In the connected part of the blockmodel, politicians have been active for a longer time, including members of NSi and former SAB and LMŠ, but without members of Svoboda.
- Connections within and between the first five groups are mainly due to co-presence in the previous National Assemblies and the state administration.



... Some conclusions

- The members of Svoboda mainly consist of two clusters (C7, C8) and, together with Levica, the cluster C6. These clusters are not co-present with each other, their members are co-present only within their own cluster.
- As expected, politicians within the three Svoboda clusters were co-present before the last parliamentary elections, each of them also includes ministers and their secretaries (e.g. the seventh most connected Svoboda cluster includes the Prime Minister and the persons who were co-present with him in the municipality of Nova Gorica, at the Faculty of Electrical Engineering, and at the company GENi; in the cluster C8 there are members of Svoboda who, on average, have collaborated for a little less than 2 years and work at the ministries of finance and public administration).
- The quite large cluster C9, which has negligible mutual co-presence, consists of newcomers in politics from different parties.
- The co-presence in university education is interesting (Faculty of Electrical Engineering, Faculty of Social Sciences, Faculty of Economics, and Law) and play a strong role.